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# Simple ways to interpret effects in modeling ordinal categorical data

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We survey effect measures for models for ordinal categorical data that can be simpler to interpret than the model parameters. For describing the effect of an explanatory variable while adjusting for other explanatory variables, we present probability-based measures, including a measure of relative size and partial effect measures based on instantaneous rates of change. We also discuss summary measures of predictive power that are analogs of *R*-squared and multiple correlation for quantitative response variables. We illustrate the measures for an example and provide `R` code for implementing them.

**KEYWORDS**
cumulative link models, cumulative logits, marginal effects, multiple correlation, proportional odds, *R*-squared, stochastic ordering

## 1 | INTRODUCTION

Popular models for ordinal categorical response variables, such as models that apply link functions to cumulative probabilities, are generalized linear models that employ nonlinear link functions. As a consequence of the nonlinearity, model parameters are not as simple to interpret as slopes and correlations for ordinary linear regression. The model effect parameters relate to measures, such as odds ratios and probits, that may not be easily understood or can even be misinterpreted by non-quantitatively oriented methodologists, see, for example, Schwartz et al. (1999).

This article surveys simpler ways to interpret the effects of an explanatory variable and to summarize the model's predictive power. In Section 2, we present alternative summaries of the effect of an explanatory variable while adjusting for other explanatory variables in the model. These include simple comparisons of the probability of extreme-response outcomes at extreme

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values of an explanatory variable, measures of average rates of change of the extreme-response probabilities, and group comparisons that result directly from latent variable models that induce standard ordinal models. In Section 3, we present measures of predictive power. Of the various measures that have been proposed, no one has achieved the popularity of $R^2$ or the multiple correlation for ordinary linear models. A straightforward approach uses $R^2$ and multiple correlation measures that closely resemble those for ordinary linear models, such as those estimated for a corresponding latent variable linear model. We propose an alternative measure that seems to relate naturally to ordinal models for cumulative probabilities of the response variable.

We illustrate existing and proposed ordinal effect measures with an example and provide `R` code for the analyses. The example uses data from a study of mental health (Agresti, 2015, section 6.3.3). The model relates a four-category ordinal response variable measuring mental impairment (1 = well, 2 = mild symptom formation, 3 = moderate symptom formation, and 4 = impaired) to a binary indicator of socioeconomic status (SES, 1 = high, 0 = low) and a quantitative life-events (LE) index that is a numerical composite measure of the number and severity of important life events such as birth of child, new job, divorce, or death in family that occurred to the subject within the past 3 years. The LE index takes values on the nonnegative integers between 0 and 9, with mean 4.3 and standard deviation 2.7. The $n = 40$ observations are available at www.stat.ufl.edu/~aa/glm/data.

## 2 | ORDINAL EFFECT MEASURES FOR INDIVIDUAL EXPLANATORY VARIABLES

For an ordinal response variable $y$ with $c$ categories, we consider models in which the explanatory variables may be a mixture of quantitative and categorical variables. We denote explanatory variable values by $\boldsymbol{x} = (x_1,\ldots,x_p)^T$. In describing ways of summarizing effects for a categorical explanatory variable, we refer also to a separate indicator variable $z$ that distinguishes between two groups.

Currently, the most popular ordinal models are special cases of the *cumulative link* model

$$\text{link}[P(y \le j)] = \alpha_j - \beta z - \beta_1 x_1 - \cdots - \beta_p x_p, \qquad j = 1, \cdots, c-1, \tag{1}$$

for link functions such as the logit and probit. The nonlinear link function naturally produces effects on the link scale. For example, for cumulative logit models, $-\beta$ is the difference between logits of cumulative probabilities when $z=1$ and when $z=0$, and $-\beta_1$ is the change in the cumulative logit per each 1-unit increase in $x_1$, adjusting for the other explanatory variables. This leads to odds ratios as natural effect measures. For instance, adjusting for the other variables, $\exp(\beta_1)$ is a multiplicative effect of each 1-unit increase in $x_1$ on the cumulative odds of response $> j$ versus $\le j$, for each $j$, and $\exp(\beta)$ is the common cumulative odds ratio for comparing the groups with $z=0$ and $z=1$.

Such effect measures are not easy to interpret by scientists who need to understand the effects in more real-world terms. In addition, with nonlinear link functions, effects often behave in a way that is counterintuitive to those mainly familiar with ordinary linear models. For example, if an explanatory variable $x'$ uncorrelated with $x_1$ is added to the model, the partial effect of $x_1$ is typically different than in the model without $x'$. For categorical variables, the effect remains the same when $x_1$ and $x'$ are *conditionally independent* given $y$ rather than marginally independent; for example, see Agresti (2013, pp. 53–54, 379–380). By contrast, the partial effect would be identical in an ordinary linear model.

We next describe three types of interpretation that supplement estimated model parameter effects with simpler effects reported on the probability scale rather than on the scale of the link function. Such effects are easier to understand and are typically more stable. The summary measures discussed in this paper are intended for cases in which an explanatory variable has a monotone effect, such as when it is a main effect term in Model (1). When an effect is U-shaped, for example, these summary measures are insufficient and potentially misleading.

## 2.1 | Extreme-category range-based probability summaries

In practice with ordinal responses, special interest often focuses on the highest and lowest response categories, the most extreme outcomes. Those categories often represent a noteworthy state, such as the *best* or *worst* outcome (e.g., complete recovery vs. death). It is informative to report how probabilities in these extreme categories, $P(y = 1)$ and $P(y = c)$, change as explanatory variables change. As any explanatory variable $x_k$ increases, cumulative link models that contain solely main effects imply monotonicity in the extreme-category probabilities but not in the other probabilities.

To summarize the effect of $x_k$ on $y$, it can be useful to report the difference between the model-fitted estimate of $P(y = 1)$ and/or $P(y = c)$, at the maximum and minimum values of $x_k$, when other explanatory variables are set at particular values such as their means. For a binary variable $z$, this is a comparison of the two groups on the extreme-category probabilities. For a continuous explanatory variable, a caveat for such measures is that their relevance depends on the plausibility of $x_k$ taking extreme values when all other explanatory variables fall at their means. Also, this summary can be misleading when outliers exist on $x_k$, in which case one can instead report the estimated probabilities at more resistant quartiles. Reporting them at the upper and lower quartiles of $x_k$ summarizes the change in $P(y = 1)$ and/or $P(y = c)$ for the middle half of the observations on $x_k$.

## 2.2 | Marginal effect measures

A second type of simple summary uses the rate of change in the probability of an extreme response category, as a function of $x_k$. We explain versions of such effects in terms of the cumulative link Model (1) for $P(y \leq j)$, which generates effects for the extreme-category probability $P(y = 1)$ and also for $P(y = c)$ by reversing the response scale. For this, we express the cumulative link model as

$$F^{-1}[P(y \leq j)] = \alpha_j - \beta z - \boldsymbol{x}^T \boldsymbol{\beta}, \qquad j = 1, \ \ldots, c-1, \tag{2}$$

where $F^{-1}$ is the inverse of a standard cumulative distribution function (cdf), $\boldsymbol{x}$ is a column vector of explanatory variable values (excluding $z$), and $\boldsymbol{\beta}$ is a column vector of parameters for $\boldsymbol{x}$. Let $f(y) = \partial F(y)/\partial y$, which is the standard normal probability density function for probit models and the standard logistic probability density function for logistic models. As a function of a particular explanatory variable, the response curve for $P(y = 1)$ (or for $P(y = c)$) looks like the curve for the corresponding binary-response model with the same link function. So for these probabilities, one can directly implement rate-of-change effect measures for binary-response models.

We first construct the effect for a quantitative explanatory variable. The rate of change in $P(y = 1)$ at a particular value of $x_k$, when other explanatory variables are fixed at certain values $\boldsymbol{x}^*$, is the partial derivative of $P(y = 1)$ with respect to $x_k$,

$$\partial P(y = 1|\boldsymbol{x} = \boldsymbol{x}^*)/\partial x_k.$$

Many sources, such as Greene (2008) and Long and Freese (2014), refer to such an instantaneous effect as a *marginal effect*. This terminology is a bit misleading, as this partial derivative refers to a conditional effect of $x_k$ rather than its marginal effect, collapsing over the other explanatory variables. Some authors (e.g., Long, 1997) instead use the term *partial effect*. Because the "marginal effect" terminology seems to be much more common, especially in relevant software that we later discuss, we will use it in this article. For the cumulative link model, the marginal effect of $x_k$ on $P(y \leq j)$, and hence on $P(y = 1)$, is $-f(\alpha_j - \beta z - \boldsymbol{x}^T\boldsymbol{\beta})\beta_k$. The marginal effect of $x_k$ on $P(y = c)$ is $f(\alpha_j - \beta z - \boldsymbol{x}^T\boldsymbol{\beta})\beta_k$.

Any particular way of fixing values of the explanatory variables has its corresponding marginal effect value for $x_k$. For the logit link, such an effect for $x_k$ on $P(y = 1)$ has the expression

$$\partial P(y = 1|\boldsymbol{x} = \boldsymbol{x}^*)/\partial x_k = \beta_k P(y = 1|\boldsymbol{x} = \boldsymbol{x}^*)[1 - P(y = 1|\boldsymbol{x} = \boldsymbol{x}^*)].$$

This takes values bounded above by its highest value of $\beta_k/4$ that occurs when $P(y = 1|\boldsymbol{x} = \boldsymbol{x}^*) = 1/2$. For cumulative probit models, the highest value of this instantaneous change is $\beta_k/\sqrt{2\pi}$, also when $P(y = 1|\boldsymbol{x} = \boldsymbol{x}^*) = 1/2$. These maximum values need not be relevant, as $P(y = 1)$ and $P(y = c)$ need not be near 1/2 for most or all the data. Freese (2014, pp. 242–246) summarized alternative versions of the marginal effect. The *average marginal effect* (AME), finds the marginal effect of $x_k$ at each of the $n$ sample values of the explanatory variables, and then averages them. Alternatively, one could compute the marginal effect with every explanatory variable, including $x_k$, set at its mean. This is called the *marginal effect at the mean* (MEM). The *marginal effect at representative values* (MER) is obtained by setting all explanatory variables at values considered to be of particular interest. For instance, when we focus on the effect of $x_k$ but a group variable $z$ is especially relevant, we could find MER for subgroups; for example, find the marginal effect for $x_k$ (a) when $z = 1$ and the explanatory variables are at their means for that group and (b) when $z = 0$ and the explanatory variables are at their means for that group.

Although our focus for ordinal responses is on the extreme categories, the various marginal effects can be formed for any outcome category. For a categorical explanatory variable, for each version, one would instead use a *discrete change*, finding the change in $P(Y = 1)$ (or $P(y = c)$) for a change in an indicator variable, holding all the other variables constant. For instance, for the $n$ sample observations on $\boldsymbol{x}$, one could find the difference between $P(y = 1)$ when $z = 1$ and when $z = 0$, and average the obtained values. Likewise, one could find the difference between $P(y = 1)$ when $z = 1$ and when $z = 0$, with other explanatory variables set at their means or at representativevalues.

Long and Freese (2014, pp. 244–246) discussed factors to consider in selecting one of these measures. Overall, they recommended AME as the best summary because it averages the effects across all cases observed in the sample and thus can be interpreted as the sample average size of the marginal effect. Greene (2008, pp. 775–785) showed how to obtain standard errors for the maximum likelihood estimators of marginal effect measures. Mood (2010) pointed out that the AME has behavior reminiscent of effects in ordinary linear models, in the sense that it is roughly stable when we add an explanatory variable to the model that is uncorrelated with the variable for which we are describing the effect. This behavior does not occur for the MEM or MER or the log odds ratio. See also Long (1997, pp.71–77), Long and Freese (2014, pp. 341–351), and Sun (2015, pp. 527–531) for discussion of the various marginal effect measures.

## 2.3 | A probability summary for ordered comparison of groups

We next present an alternative way to summarize the effect of a categorical explanatory variable on an ordinal response $y$, suggested by Agresti and Kateri (2017) and developed in a more general context by Thas et al. (2012). We discuss this in the context of comparing two groups ($z = 0$ and $z = 1$).

It is often sensible to regard an ordinal categorical variable as crude measurement of an underlying continuous latent variable $y^*$ that, if we could observe it, would be the response variable in an ordinary linear model. In fact, Anderson and Philips (1981) showed that the cumulative link Model (2) is implied by a model in which a latent response has conditional distribution with standard cdf given by the inverse of the link function. Let $y_1^*$ and $y_2^*$ denote independent underlying latent variables for the ordinal categorical response, representing the underlying distributions when $z = 1$ and when $z = 0$, respectively. At a particular setting $\boldsymbol{x}$ for other explanatory variables, $P(y_1^* > y_2^* | \boldsymbol{x})$ is a summary measure of relative size. This measure is most meaningful when the groups are stochastically ordered, such as when they differ by a location shift on some scale, and it is sometimes referred to as a measure of *stochastic superiority*.

The normal latent variable model with $y^* \sim N(\beta z + \beta_1 x_1 + \cdots + \beta_p x_p, 1)$ implies the cumulative probit model

$$\Phi^{-1}[P(y \le j)] = \alpha_j - \beta z - \beta_1 x_1 - \cdots - \beta_p x_p,$$

with $\{\alpha_j\}$ being cutpoints on the underlying scale and $\Phi$ being the standard normal cdf. For this model,

$$P(y_1^* > y_2^* | \boldsymbol{x}) = P\left[\frac{(y_1^* - y_2^*) - \beta}{\sqrt{2}} > \frac{-\beta}{\sqrt{2}}\right] = \Phi\left(\frac{\beta}{\sqrt{2}}\right). \tag{3}$$

This is true regardless of the $\boldsymbol{x}$ value, so we simplify the notation to $P(y_1^* > y_2^*)$. For the logit link, Agresti and Kateri (2017) showed that

$$P(y_1^* > y_2^*) \approx \frac{\exp(\beta/\sqrt{2})}{[1 + \exp(\beta/\sqrt{2})]}, \tag{4}$$

for the $\beta$ coefficient of $z$ in the cumulative logit model. For a log-log link, which is relevant when we expect underlying latent variables to have extreme-value distributions, Agresti and Kateri noted that

$$P(y_1^* > y_2^*) = \frac{\exp(\beta)}{[1 + \exp(\beta)]},$$

for the $\beta$ coefficient of $z$ in the cumulative link model with log-log link. Ordinary confidence intervals for the $\beta$ model parameter induce confidence intervals for the stochastic superiority measure.

Agresti and Kateri suggested that many practitioners can more easily interpret $P(y_1^* > y_2^*)$ than parameters such as odds ratios and differences in probits that naturally result in cumulative link models. They also proposed related measures for the observed $y$ scale that need not relate to latent variables.

The ordinal effect measures presented in this section extend directly to summary comparisons of multiple groups, based on more general models that have multiple indicator variables for the groups. For example, suppose a cumulative probit model contains terms $\beta^{(a)} z_a + \beta^{(b)} z_b$ in the linear predictor for groups $a$ and $b$, where $z_j = 1$ for observations from group $j$ and $z_j = 0$ otherwise. Then, an analog of (3) for comparing those groups is $\Phi[(\beta^{(a)} - \beta^{(b)})/\sqrt{2}]$.

Such probability measures also generalize for some more complex models, such as the cumulative link mixed model that has a subject-specific random intercept. If a group comparison $\beta$ refers to a within-subject effect, then with the probit link, $P(y_1^* > y_2^*) = \Phi(\beta/\sqrt{2})$, with the corresponding approximation for the logit link.

## 2.4 | Example: Effect measures for individual explanatory variables

We now illustrate the effect measures for the data from the study of mental health mentioned in the Section 1. Different packages in R permit fitting cumulative link models, but none of them implement many of the measures we have described. Separate functions are available for some measures, and we developed new functions based on existing ones. For fitting cumulative link models, we used the *polr* function of the R–package MASS. We illustrate with the cumulative logit model implied by the logistic latent variable linear model. For the data set on mental health, the maximum likelihood fit for modeling $y$ (mental impairment) is

$$logit[\hat{P}(y \leq j)] = \widehat{\alpha}_j + 1.111(\text{SES}) - 0.319(\text{LE}).$$

Table 1 shows model fitting and results, with edited output.

For a quantitative variable, such as LE in the mental impairment data set, we can report the change in an extreme-category probability over its range, at the means of other explanatory variables or at particular categories of qualitative explanatory variables. Table 2 shows how to obtain the estimated changes when LE changes from its minimum to its maximum value, separately for low SES and high SES subjects. For either SES group, as LE increases, the probability decreases substantially for the *well* category (by 0.389 for low SES and by 0.581 for high SES) and increases substantially for the *impaired* category (by 0.560 for low SES and by 0.354 for high SES). These changes characterize in a simple manner the very strong effect of

**TABLE 1** R code and output (edited) for the cumulative logit model fitted to the mental impairment data

```
> Mental <- read.table("http://www.stat.ufl.edu/~aa/glm/data/Mental.dat",header=T)
> head(Mental) # the first 4 of the 40 observations
  impair ses life
1      1   1    1
2      1   1    9
3      1   1    0
4      1   1    4
...
> attach(Mental)
> library(MASS)
> impair.f <- factor(impair) # polr requires response to be a factor
> logit.m <- polr(impair.f ~ ses + life, method="logistic")
> summary(logit.m)
Coefficients:
       Value Std. Error t value # reported t actually is a z Wald statistic
ses  -1.1112     0.6109  -1.819
life  0.3189     0.1210   2.635
```

**TABLE 2** R code and output (edited) for extreme-category probability changes in cumulative logit model for mental impairment

```
> pred_max0 <- predict(logit.m, data.frame(ses=0,life=max(life)), type="probs")
> pred_max0
         1          2          3          4 # predicted outcome prob's
0.04102612 0.11914448 0.18048372 0.65934567
> pred_min0 <- predict(logit.m, data.frame(ses=0,life=min(life)), type="probs")
> pred_min0
         1          2          3          4
0.42998727 0.34080542 0.13029529 0.09891202
> pred_max0[c(1,4)] - pred_min0[c(1,4)]
         1          4
-0.3889612  0.5604337 # LE effect (at max - at min) in cat's 1 and 4 when SES=0.

> pred_max1 <- predict(logit.m, data.frame(ses=1,life=max(life)), type="probs")
> pred_min1 <- predict(logit.m, data.frame(ses=1,life=min(life)), type="probs")
> pred_max1[c(1,4)] - pred_min1[c(1,4)]
         1          4
-0.5811892  0.3542873  # LE effect (at max - at min) in cat's 1 and 4 when SES=1.

> pred1 <- predict(logit.m, data.frame(ses=1,life=mean(life)), type="probs")
> pred0 <- predict(logit.m, data.frame(ses=0,life=mean(life)), type="probs")
> pred1[c(1,4)] - pred0[c(1,4)]
         1          4
 0.2078490 -0.1764812 # these are discrete marginal effects of SES at mean of LE
```

*Note.* The changes compare the maximum and minimum life events values, at low SES and at high SES, and compare low and high SES at the mean for life events. SES = socioeconomic status; LE = life-events.

LE on mental impairment. Table 2 also shows the estimated changes between the SES levels, at the mean of LE, which is a discrete-change version of the MEM. For high SES compared with low SES at the mean of LE, the estimated probability is 0.208 higher for the *well* category and 0.176 lower for the *impaired* category.

We next consider marginal effects. The `R`–package erer of Sun (2016) has a function *ocME* that supplies marginal effects at the mean, using output from the *polr* function. Table 3 shows results, focusing again on the extreme response categories. At the mean of LE, the rate of change in the estimated probability per unit change in LE is −0.062 for the *well* outcome and 0.049 for the *impaired* outcome. For categorical explanatory variables, it reports the discrete change. When SES increases from 0 to 1 at the mean of LE, the estimated probability of the *well* outcome increases by 0.208 and the estimated probability of the *impaired* outcome decreases by 0.176. These are the same measures we just found and reported at the bottom of Table 2. The *ocME* function employs only logit and probit link functions. An extension of it (called ocMEM) that handles also log-log and complementary log-log link functions is available from the authors and at the Supporting Information available at www.stat.ufl.edu/~aa/articles/agresti_tarantola_appendix.pdf.

The `erer` package does not report AMEs, so we constructed a function called *ocAME* based on the *ocME* function that handles also log-log and complementary log-log link functions. The

**TABLE 3** R code and output (edited) for marginal effect at the mean and average marginal effect for the cumulative logit model fitted to the mental impairment data

```
> library(erer)
> ocME(logit.m) # for marginal effects at the mean
     effect.1 effect.2 effect.3 effect.4
ses     0.208    0.053   -0.084   -0.176
life   -0.062   -0.014    0.027    0.049

> ocAME(logit.m) # new function available from the authors

$ME.1 # category 1 (well)
     effect std.error z.value p.value
ses   0.198     0.104   1.913   0.056
life -0.057     0.019  -3.005   0.003

$ME.4 # category 4 (impaired)
     effect std.error z.value p.value
ses  -0.171     0.094  -1.819   0.069
life  0.048     0.017   2.780   0.005
```

function, available at the online site just mentioned, uses the discrete-change version when an explanatory variable is categorical. Table 3 also shows results of applying this function, for the extreme categories. At the 40 observed values for LE and SES, the rate of change in the estimated probability per unit change in LE averages to −0.057 for the *well* outcome and to 0.048 for the *impaired* outcome. At the 40 observed values for LE, when SES increases from 0 to 1, the estimated probability of the *well* outcome increases by an average of 0.198 and the estimated probability of the *impaired* outcome decreases by an average of 0.171.

Finally, we estimate the ordinal comparison measure introduced in Section 2.3, by comparing the SES groups while adjusting for LE. An exact estimate using Formula 3 follows directly from the SES effect estimate in the cumulative probit model. Table 4 shows its value of 0.314 and its 95% profile likelihood confidence interval. For the cumulative logit model, we can use the approximate Formula 4, which gives a similar result (here, 0.313) because of the similarity of logit and probit link functions. So the estimated probability is 0.31 that mental impairment is worse at high SES than at low SES, adjusting for the LE index.

## 3 | SUMMARY MEASURES OF PREDICTIVE POWER

Next, we discuss ways to summarize how well we can predict $y$ using the fit of the chosen ordinal model, as described by the explanatory power of the explanatory variables. Such measures can be useful for comparing different models, such as to see whether it helps substantively to add an interaction term. A model that is more complex than a working model need not provide much more explanatory power, regardless of whether its extra terms are statistically significant. Here, by explanatory power, we mean something distinct from goodness of fit. A model may fit a particular data set very well even if the explanatory power that the model provides is small.

**TABLE 4** R code and output for stochastic superiority comparison of SES groups, using cumulative probit and logit models fitted to the mental impairment data

```
> probit.m <- polr(impair.f ~ ses + life, method = "probit")
> summary(probit.m) # we don't show intercept parameter estimates
Coefficients:
       Value Std. Error t value
ses  -0.6834    0.36411  -1.877
life  0.1954    0.06887   2.837

> pnorm(probit.m$coefficients[1]/sqrt(2)) # using formula 3
      ses
0.3144742 # ML estimate of probit ordinal comparison measure for SES
> pnorm(confint(probit.m)[1,]/sqrt(2))
    2.5 %    97.5 %  # profile likelihood CI
0.1608011 0.5074903

> exp(logit.m$coefficients[1]/sqrt(2))/(1+exp(logit.m$coefficients[1]/sqrt(2)))
    ses
0.31308 # formula (4) approx ordinal comparison for logit link
```

*Note.* SES = socioeconomic status.

## 3.1 | Concordance index

The *concordance index* (Harrell et al. 1996, section 5.5) is a measure of predictive power that strictly uses ordinality and is natural for models that imply stochastic orderings at various settings of the explanatory variables. Consider all pairs of observations that have different outcomes on $y$. The concordance index estimates the probability that the predictions and the outcomes are concordant, that is, that the observation with the larger $y$-value also has a stochastically higher set of model-fitted probabilities. For cumulative link models, the stochastic ordering of the model-fitted probabilities is identical to the ordering of linear predictor values without the intercept. Of those pairs that are untied on $y$ but tied on the linear predictor, half are treated as concordant and half as discordant, so that the concordance index has a null value of 1/2. The higher the value above 1/2, the better the predictive power.

The concordance index is a linear transformation to the [0, 1] scale of a version of the ordinal measure of association called *Somers' d*. That measure, which falls on the [−1, 1] scale, is the difference between the proportions of concordant and discordant pairs out of those pairs that are untied on $y$. That is, for marginal response counts $n_j$, $j = 1, \ldots, c$, with $n = \sum_j n_j$ and $C$ concordant pairs and $D$ discordant pairs, this version of the Somers measure is

$$d = \frac{C - D}{\left[\frac{n(n-1)}{2} - T_y\right]},$$

where $T_y = \sum_j n_j(n_j - 1)/2$ denotes the number of pairs that are tied on $y$. The concordance index equals $(d+1)/2$.

Appealing features of the concordance index are its simple structure and its generality of potential application. Because it utilizes ranking information only, however, it cannot

distinguish between different link functions or linear predictors that yield the same stochastic orderings. With a single linear predictor in a cumulative link model, for instance, the concordance index assumes the same value for logit and complementary log-log link functions, even though the model fits can be quite different.

## 3.2 | *R*-squared type measures

An alternative approach to summarizing predictive power adapts standard measures for quantitative response variables. For example, to mimic $R^2$ for ordinary linear models, we could assign ordered scores $\{v_j\}$ to the categories of $y$ and find the proportional reduction in variance in comparing the marginal variation to the conditional variation (Agresti, 1986). That measure has the disadvantage of requiring response scores, which cumulative link models do not require. A way to construct a measure without assigning scores is to estimate $R^2$ for the linear model for an underlying latent response variable. McKelvey and Zavoina (1975) suggested this measure for the cumulative probit model, for which the underlying latent variable model is the ordinary normal linear model. Let $y_i^*$ denote the value of the latent variable for subject $i$. The $R^2$ measure has the usual proportional reduction in variation form

$$R^2 = \frac{\sum_i (y_i^* - \bar{y}^*)^2 - \sum_i (y_i^* - \hat{y}_i^*)^2}{\sum_i (y_i^* - \bar{y}^*)^2} = \frac{\sum_i (\hat{y}_i^* - \bar{y}^*)^2}{\sum_i (y_i^* - \bar{y}^*)^2}.$$

This equals the estimated variance of $\hat{y}^*$ divided by the estimated variance of $y^*$. After fitting a cumulative link model, we can estimate the variance of $\hat{y}^*$ by the variance of the linear predictor $\hat{y}^* = \hat{\beta}z + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p$ without the intercepts. We cannot observe the latent variable or its sample variance, but we can estimate that variance by the estimated variance of $\hat{y}^*$ plus the variance of the latent variable distribution, which is 1 for the probit link and $\pi^2/3$=3.29 for the logit link (i.e., standard logistic distribution).

An alternative proportional-reduction-in-variability approach uses a likelihood-based measure such as was proposed for binary data by McFadden (2014). We can express this in terms of deviance measures for the ungrouped data file. Denote the residual deviance by $D_M$ for the working model fit and denote the null deviance (i.e., for the model containing only intercept terms) by $D_0$ . Denote the corresponding maximized log-likelihood values by $L_M$ and $L_0$. The *pseudo R-squared* measure

$$\frac{D_0 - D_M}{D_0} = 1 - \frac{L_M}{L_0},$$

equals 0 when the model provides no improvement in fit over the null model and it equals 1 when the model fits as well as the saturated model. A weakness of such a measure and related ones based on the log-likelihood is that the scale is not the same as for $y$. Interpreting the numerical value is difficult, other than in a comparative sense for different models.

For surveys of $R^2$ type measures in various contexts (but mainly for binary responses), see Liao and McGee (2003), Mittlböck and Schemper (1996), and Zheng and Agresti (2000).

## 3.3 | Multiple correlation measures

Some statisticians prefer correlation measures over related $R^2$ measures because of the appeal of working on the original scale and its proportionality to the effect size. For example, for the ordinary linear model, for fixed marginal standard deviations, doubling the slope also doubles the correlation.

For ordinal modeling, we could estimate the multiple correlation for the underlying latent variable model, using the square root of the McKelvey and Zavoina (1975) $R^2$. As an alternative, here, we propose an approach that does not require reference to a latent variable or assigning arbitrary scores to $y$. We use as scores the average cumulative proportions for the marginal distribution of $y$ because of their natural connection with cumulative link models. For sample marginal proportions $\{p_j\}$, the average cumulative proportion in category $j$ is

$$v_j = \sum_{k=1}^{j-1} p_k + \left(\frac{1}{2}\right) p_j, \quad j = 1, 2, \ \ldots, c.$$

Such scores, which are linearly related to the midranks $\{r_j\}$ by

$$r_j = nv_j + 0.5, \quad v_j = (r_j - 0.5)/n,$$

are sometimes referred to as *ridits*. See Agresti (2010, section 2.1) for discussion and examples of their use. In particular, (a) they satisfy $\sum_{j=1}^{c} p_j v_j = 0.50$, (b) if two adjacent categories of $y$ are combined, then the ridit score for the new category falls between the original two scores, with the other scores being unaffected, and (c) if the category ordering is reversed, the ridit score for category $j$ transforms from $v_j$ to $(1-v_j)$. With such scores, we construct the correlation for the $n$ sample observations between the observed outcome category score for a subject and the estimated mean score generated by the model-fitted probability values for the subject. With ridit or midrank scores, this is a multiple correlation version of the Spearman correlation. Such scores are especially natural for the cumulative logit model because McCullagh (1980) showed that the components of the efficient score are cross-products of the explanatory variables with the average rank for the response category. (For example, for comparing two groups with that model, the score test is identical to the two-sample Wilcoxon test.)

Yet another way to circumvent assigning scores is to treat them as parameters. For example, one could estimate scores for the outcome categories for which the correlation is maximized between them and the fitted mean score. This is a type of canonical correlation as a multiple correlation. One could also consider the special case of this approach in which the parameter scores are restricted to be monotone increasing.

Zheng and Agresti (2000) proposed a multiple correlation measure for generalized linear models. For its application to binary regression models, they found that a jackknife estimate is less biased. It would be of interest to study whether this is also true for ordinal versions of the measure such as the rank-based measure just proposed.

## 3.4 | Example: Measures of predictive power

We now illustrate the measures of predictive power for the data from the study of mental health. The concordance index can be easily obtained in `R` using an `R` package that has a function for Somers' $d$, as shown in Table 5. The command *logit.m$lp* provides the fitted values of the linear predictor without the intercepts. For the cumulative logit model, we estimate that for 70.5% of

**TABLE 5** R code and output (edited) for concordance index for cumulative logit model with mental impairment data

```
> logit.m <- polr(impair.f ~ ses + life, method="logistic")
> library(DescTools)
> (SomersDelta(logit.m$lp, impair) + 1)/2
[1] 0.7047377
```

the untied pairs on mental impairment, the observation with the higher mental impairment also had a stochastically higher estimated distribution.

Table 6 shows how to find various $R^2$ measures. The McFadden pseudo $R^2$ measure is easily calculated using deviances or maximized log-likelihoods. The deviance for the model is 9.1% smaller than for the null model. For the logistic latent variable model, the multiple correlation is 0.473, with $R^2 = 0.224$. We estimate that for the underlying continuous measure of mental impairment, the conditional variability (given LE and SES) is 22.4% less than the marginal variability. For the 40 observations, with ridit scores (which are 0.15, 0.45, 0.6875, 0.8875), the multiple correlation is 0.479 and $R^2 = 0.230$. The $R^2$ values with the latent variable model and

**TABLE 6** R code and output (edited) for $R^2$ and multiple correlation measures for cumulative logit model with mental impairment data

```
> logit.m <- polr(impair.f ~ ses + life, method="logistic")
> logit.m0<- polr(impair.f ~ 1, method="logistic")

> R2 <- (logit.m0$deviance - logit.m$deviance)/logit.m0$deviance
> R2  # McFadden pseudo R-squared
[1] 0.09119561
1 - logLik(logit.m)/logLik(logit.m0)
'log Lik.' 0.09119561 (df=5)

> var(logit.m$lp)/(var(logit.m$lp) + (pi^2)/3)
[1] 0.2237609 #  R-squared for latent var. model for cumulative logit
> sqrt(0.2237609)
[1] 0.4730337 # multiple correlation for latent variable model

> pred <- predict(logit.m, type = "probs")
> ridits <- (rank(impair) - 0.5)/40; ridits # rank fn. gives midrank scores
 [1] 0.1500 0.1500 0.1500 0.1500 0.1500 0.1500 0.1500 0.1500 0.1500 0.1500 0.1500
[12] 0.1500 0.4500 0.4500 0.4500 0.4500 0.4500 0.4500 0.4500 0.4500 0.4500 0.4500
[23] 0.4500 0.4500 0.6875 0.6875 0.6875 0.6875 0.6875 0.6875 0.6875 0.8875 0.8875
[34] 0.8875 0.8875 0.8875 0.8875 0.8875 0.8875 0.8875

> pred.ridit <- 0.15*pred[,1] + 0.45*pred[,2] + 0.6875*pred[,3] + 0.8875*pred[,4]
>  cor(ridits, pred.ridit); cor(ridits, pred.ridit)^2
[1] 0.4793919 # Spearman multiple correlation analog
[1] 0.2298166 # an R-squared for rank scores
```

with ridit scores are similar and seem to be more realistic summaries of predictive power for these data than the pseudo $R^2$ of 0.091 provides.

## 4 | EXTENSIONS TO OTHER GENERALIZED LINEAR MODELS

The focus of this paper has been ordinal response variables. The same issues arise for other types of response variables and model structures. For instance, one could develop alternative effect measures and measures of predictive power for nominal-scale variables and for marginal models and random-effects models.

The measures discussed in this paper extend also to more general ordinal-response models than those having linear predictors, such as generalized additive models for ordinal responses and their extensions with random effects, see, for example, Yee and Wild (1996) and Wood et al. (2016). Readers who find it challenging to understand cumulative link models and their corresponding summary measures such as odds ratios undoubtedly find such generalized models even more demanding. When effects are monotone, simple summaries such as changes over the range in estimated ordinal extreme-category probabilities could be useful to help less quantitatively sophisticated readers understand the substantive importance of the effects, and they can be presented with the model summaries (e.g., as done by Wood, 2016) with a graphic in their example on prostate cancer). Interesting challenges for research statisticians are abundant for such models, such as obtaining confidence intervals for some of the summary measures that can be applied with such models, such as $P(y_1^* > y_2^* | \boldsymbol{x})$ and differences of extreme-category probabilities for comparing two groups.

In addition, there is scope to develop simple summary measures for alternative generalized linear models that employ nonlinear link functions and may have nontrivial interpretations. For example, the measure $P(y_1 > y_2; \boldsymbol{x})$ could, in principle, be used with models for any quantitative response, such as for comparing groups in gamma-regression models for positive responses and in standard survival models for continuous responses, such as proportional hazards models. With the usual adjustment for ties, they could also be used with Poisson, negative binomial, and zero-inflated models for count-data responses. It is of interest to consider such cases and investigate whether the value of the measure is still relatively unaffected by the values of the explanatory variables.

Finally, although our focus has been on simple ways of describing effects, of course, other measures are still highly relevant in any model-building process. An example is Akaike information criterion, useful for comparing models in terms of estimating which is likely to give fitted values closest to the underlying probabilities or means.

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